

National University of Computer and Emerging Sciences, Lahore Campus

	Lab Name:	Artificial Intelligence	Course Code:	
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	
	Paper Date:		Section	4F
	Exam Type:	Quiz 1	Page(s):	

Student Name

Roll No.

Section:

Quiz 1

Q: What will be the output of the following code? [2 marks]

```
lst = [3, 1, 4, 2]
for i in range(1, len(lst)):
    if lst[i] < lst[i-1]:
        lst[i] = lst[i-1]
print(lst)
```

Step-by-step:

- $i=1 \rightarrow 1 < 3 \rightarrow \text{replace} \rightarrow [3, 3, 4, 2]$
- $i=2 \rightarrow 4 < 3 \rightarrow \text{no change} \rightarrow [3, 3, 4, 2]$
- $i=3 \rightarrow 2 < 4 \rightarrow \text{replace} \rightarrow [3, 3, 4, 4]$

Final Output:

[3, 3, 4, 4]

Q: [4 marks]

Python Statement	Valid/Syntax Error
result = [x, y for x in range(3) for y in range(2)]	Syntax Error (incorrect list comprehension)
print(bool([]), bool({}), bool(()))	Valid → Output: False False False
my_tuple = (1, 2, 3) my_tuple.append(4)	Error (tuples are immutable, no append)

```
x = input() print(x  
* 2)
```

Error (tuples are immutable, no append)

Q: Write a function digitProduct(n) that returns the product of digits of a positive integer. [4 marks]

Example: Input: n = 243, Output: 24

```
def digitProduct(n):  
    product = 1  
    while n > 0:  
        digit = n % 10  
        product *= digit  
        n //= 10  
    return product
```

Q: Write Python code using Pandas to: [5 marks]

- ***Create a DataFrame from the following data***

```
data = { "Name": ["Ali", "Sara", "Hina", "Omar"],  
        "Department": ["HR", "IT", "IT", "Finance"],  
        "Salary": [45000, 70000, 65000, 50000] }
```

- ***Display employees working in the IT department***
- ***Find the average salary***

```
import pandas as pd
```

```
# Create DataFrame
```

```
data = {  
  
    "Name": ["Ali", "Sara", "Hina", "Omar"],
```

```
"Department": ["HR", "IT", "IT", "Finance"],  
"Salary": [45000, 70000, 65000, 50000]  
}
```

```
df = pd.DataFrame(data)
```

```
# Employees in IT department
```

```
it_employees = df[df["Department"] == "IT"]
```

```
print(it_employees)
```

```
# Average salary
```

```
avg_salary = df["Salary"].mean()
```

```
print("Average Salary:", avg_salary)
```